Risk factors and preventive interventions for Alzheimer disease: state of the science.

BACKGROUND: Numerous studies have investigated risk factors for Alzheimer disease (AD). However, at a recent National Institutes of Health State-of-the-Science Conference, an independent panel found insufficient evidence to support the association of any modifiable factor with risk of cognitive decline or AD. OBJECTIVE: To present key findings for selected factors and AD risk that led the panel to their conclusion. DATA SOURCES: An evidence report was commissioned by the Agency for Healthcare Research and Quality. It included English-language publications in MEDLINE and the Cochrane Database of Systematic Reviews from 1984 through October 27, 2009. Expert presentations and public discussions were considered. STUDY SELECTION: Study inclusion criteria for the evidence report were participants aged 50 years and older from general populations in developed countries; minimum sample sizes of 300 for cohort studies and 50 for randomized controlled trials; at least 2 years between exposure and outcome assessment; and use of well-accepted diagnostic criteria for AD. DATA EXTRACTION: Included studies were evaluated for eligibility and data were abstracted. Quality of overall evidence for each factor was summarized as low, moderate, or high. DATA SYNTHESIS: Diabetes mellitus, hyperlipidemia in midlife, and current tobacco use were associated with increased risk of AD, and Mediterranean-type diet, folic acid intake, low or moderate alcohol intake, cognitive activities, and physical activity were associated with decreased risk. The quality of evidence was low for all of these associations. CONCLUSION: Currently, insufficient evidence exists to draw firm conclusions on the association of any modifiable factors with risk of AD.